

ESKISEHIR HAVALIMANI, Turkey

WMO: 171240

Lat: 39.781N

Lon: 30.580E

Elev: 786

StdP: 92.23

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-7.8	-12.7	1.4	-6.7	-10.2	1.7	-5.6	9.7	1.7	8.5	1.4	2.0	90	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	33.5	18.9	31.9	18.5	30.2	18.0	21.3	29.6	20.4	28.3	19.6	27.3	3.6	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.9	25.1	17.7	14.0	24.5	16.8	13.2	23.6	65.8	29.0	62.1	28.4	59.3	27.2	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.6	-14.0	36.4	3.5	1.6	-16.6	37.5	-18.7	38.5	-20.7	39.4	-23.2	40.6		
(5)			-14.5	22.7	3.5	2.2	-17.0	24.3	-19.0	25.6	-20.9	26.8	-23.4	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	0.7	2.6	6.2	10.7	15.6	19.5	22.5	22.6	18.2	12.8	6.8	2.2		
	DBStd	8.48	4.62	4.59	4.07	3.93	3.28	2.75	2.33	2.39	3.03	3.67	4.07	4.24		
	HDD10.0	1032	288	208	128	37	2	0	0	0	0	16	111	243		
	HDD18.3	2766	546	440	375	228	94	19	1	2	39	176	346	500		
	CDD10.0	1673	0	2	11	59	176	285	387	390	246	102	14	2		
	CDD18.3	364	0	0	0	1	11	54	129	133	34	2	0	0		
	CDH23.3	4418	0	0	2	37	221	693	1463	1427	512	63	0	0		
	CDH26.7	1641	0	0	0	4	41	223	611	592	162	9	0	0		
(14) Wind	WSAvg	2.9	3.0	3.0	3.2	3.0	2.7	2.9	3.4	3.2	2.7	2.3	2.4	2.8		
(15) Precipitation	PrecAvg	390	43	31	40	41	46	35	12	11	15	31	30	47		
	PrecMax	507	104	78	79	117	127	128	64	40	83	98	73	128		
	PrecMin	296	3	2	7	9	8	2	0	0	0	1	1	1		
	PrecStd	57	27	20	19	22	28	28	14	11	17	20	17	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	17.5	22.1	26.9	30.0	33.2	36.1	35.2	33.1	28.1	20.8	14.9		
		MCWB	8.3	10.1	10.7	14.5	15.9	18.5	20.0	19.3	17.9	17.3	12.1	10.1		
	2%	DB	10.9	14.1	18.9	24.0	27.8	31.0	33.8	33.4	30.6	25.0	18.1	12.0		
		MCWB	7.2	8.3	10.2	13.3	15.8	18.5	19.2	18.8	17.0	15.3	11.6	8.7		
	5%	DB	8.8	12.0	16.5	21.5	26.0	29.2	31.9	31.8	28.3	22.8	15.8	10.0		
		MCWB	6.1	7.5	9.1	12.2	15.6	17.6	18.9	18.5	16.5	14.1	10.5	7.4		
	10%	DB	6.9	9.8	14.1	19.1	23.9	27.5	30.1	30.1	26.2	20.2	13.8	8.1		
		MCWB	5.1	6.3	8.0	11.3	14.6	17.1	18.3	18.3	16.0	13.2	9.5	6.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	10.8	12.5	16.0	19.2	21.1	23.1	22.6	20.5	19.1	13.6	10.6		
		MCDB	12.6	15.7	19.7	24.0	27.2	29.2	32.3	31.9	28.8	25.7	18.0	13.8		
	2%	WB	7.5	8.9	10.8	14.1	17.6	19.8	21.5	21.3	19.1	16.3	12.1	8.9		
		MCDB	10.4	13.4	17.3	22.2	25.3	28.3	30.2	29.5	27.0	21.8	16.8	11.6		
	5%	WB	6.2	7.7	9.6	13.0	16.4	18.7	20.5	20.3	18.0	15.0	10.9	7.8		
		MCDB	8.3	11.5	15.5	20.5	23.4	27.1	28.5	28.0	25.5	20.3	14.9	9.6		
	10%	WB	5.0	6.4	8.4	11.7	15.3	17.8	19.5	19.5	17.0	14.0	9.8	6.5		
		MCDB	6.6	9.4	13.5	17.9	21.9	25.5	27.2	26.9	23.9	19.0	13.3	8.1		
(35) Mean Daily Temperature Range	MDBR		6.8	9.3	11.7	13.0	13.7	14.1	14.8	14.7	15.0	13.0	10.9	7.0		
	5% DB	MCDBR	9.7	13.5	16.8	17.7	17.2	17.2	17.7	17.3	18.8	17.2	14.4	8.9		
		MCWBR	6.8	8.5	9.4	8.7	7.6	6.8	6.3	5.6	7.5	8.5	8.6	6.0		
	5% WB	MCDBR	8.9	12.4	14.8	15.6	15.0	15.5	15.1	14.5	16.1	14.0	12.4	7.9		
		MCWBR	6.5	8.1	8.7	8.3	7.7	7.2	6.8	5.9	7.7	7.9	7.8	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.341	0.387	0.425	0.404	0.383	0.380	0.382	0.366	0.358	0.326	0.314		
	taud		2.383	2.319	2.199	2.122	2.252	2.342	2.354	2.338	2.374	2.387	2.437	2.428		
	Ebn at Noon		847	870	861	850	875	892	891	882	875	841	828	819		
	Edh at Noon		85	105	131	151	136	124	122	121	109	97	80	76		
(44) All-Sky Solar Radiation	RadAvg		1.83	2.71	3.86	5.00	6.01	6.94	7.36	6.47	5.02	3.43	2.37	1.63		
	RadStd		0.20	0.35	0.34	0.49	0.48	0.61	0.28	0.30	0.37	0.26	0.28	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-1.24	-1.02	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page